Ports

Think of your computer or phone as a massive international hotel.

The IP Address is the street address of the hotel. It gets the data to the right building. But once the data arrives at the building, it needs to know which room to go to. Those "room numbers" are the Ports.

Why do we need Ports?

Without ports, your computer could only do one thing at a time. If you were downloading a file, you couldn't browse the web because the computer wouldn't know which data belonged to the download and which belonged to the browser.

Ports allow your operating system to sort incoming data packets and deliver them to the correct application.

How Ports Work (The 5-Tuple)

To make a successful connection, your device uses a "5-Tuple" to ensure data doesn't get lost. It's like a digital shipping label:

Source IP: Your phone's address.

Source Port: A random room number your phone picked to send the data from.

Destination IP: The Telegram server's address.

Destination Port: The specific "room" the Telegram server is listening to (usually Port 443).

Protocol: Whether it's using TCP (reliable) or UDP (fast).

In the OSI model we discussed, TCP and UDP live at Layer 4 (Transport Layer). If the IP Address is the address of a building (your phone or a server), then Ports are the specific apartment numbers inside that building.

Ports allow a single device to run multiple applications at once without the data getting mixed up. For example, your phone can handle a Telegram message, a Spotify stream, and a Google Chrome tab all at the same time because each uses a different port.

TCP (Transmission Control Protocol)

TCP is like a registered letter. It is "connection-oriented," meaning it establishes a formal connection before sending data and checks to make sure every single piece arrived in the correct order.

How it works: It uses a "Three-Way Handshake" (Synchronize -> Synchronize/Acknowledge -> Acknowledge).

Best for: Tasks where accuracy is more important than speed (e.g., sending a text message on Telegram, downloading a file, or loading a webpage).

If a packet is lost: TCP notices it’s missing and asks the sender to send it again.

UDP (User Datagram Protocol)

UDP is like regular mail or a shouting into a megaphone. It is "connectionless," meaning it just sends the data as fast as possible without checking if anyone is listening or if the data arrived safely.

How it works: It just pours data out. There is no handshake and no recovery for lost data.

Best for: Tasks where speed is the only thing that matters (e.g., a Telegram Voice Call or a Video Stream).

If a packet is lost: You might hear a tiny "glitch" or "pop" in the audio, but the call keeps going. You don't want the audio to "pause" just to recover a missing millisecond of sound from 10 seconds ago.

Common Port Numbers

There are 65,535 available ports. They are generally categorized like this:

Port RangeCategoryDescription0 – 1,023Well-Known PortsReserved for standard services (HTTP, FTP, etc.)1,024 – 49,151Registered PortsUsed by companies for specific apps (like Telegram).49,152 – 65,535Dynamic/PrivateTemporary ports your phone uses to talk to servers.

Real-World Examples:

Port 80 (TCP): Standard unencrypted web traffic (HTTP).

Port 443 (TCP): Secure web traffic (HTTPS)—this is what Telegram uses to talk to its servers.

Port 53 (UDP/TCP): DNS (The system that turns "https://www.google.com/search?q=google.com" into an IP address).

Port 5222 (TCP): Often used by messaging apps for client-to-server communication.

How they work together in Telegram

When you use Telegram, your phone actually uses both depending on what you are doing:

Sending a Photo (TCP): Telegram uses a TCP port to ensure every pixel of that photo arrives perfectly. If the connection is shaky, it retries until the file is 100% there.

A Video Call (UDP): Telegram switches to a UDP port. It wants the lowest possible "lag" (latency). If a few frames of video are lost due to bad Wi-Fi, the app just skips them to keep the video "live."

ports and theit uses with the attacks they are vulnerable for

1. File Transfer & Remote Access

These ports allow users to move data or control a computer from afar. Because they provide direct access to the system, they are primary targets for hackers.

Port 20 (FTP Data): Used to transfer actual file data in "Active Mode." It is vulnerable to Clear-text sniffing because the data is unencrypted.

Port 21 (FTP Control): Used for sending commands like login and file listing. It is a major target for Brute-force attacks and Anonymous login exploits.

Port 22 (SSH): Provides secure, encrypted remote login. While safe, it is frequently targeted by Credential stuffing and automated Brute-force bots.

Port 23 (Telnet): An older, unencrypted remote login method. It is highly dangerous because of Packet sniffing, where passwords are sent in plain text.

Port 69 (TFTP): A simple file transfer protocol that requires no login. Its main vulnerability is Unauthorized access since it lacks authentication.

Port 3389 (RDP): Windows Remote Desktop. It is famous for the BlueKeep vulnerability and is a top target for RDP Hijacking.

Port 5900 (VNC): Used for remote desktop sharing. It is vulnerable to Weak password cracking and often lacks strong default encryption.

Port 2049 (NFS): Used for sharing files across a network. It can suffer from Information leakage

and unauthorized file mounting if not configured correctly.

2. Web & Infrastructure Services

These ports are the "front door" of the internet. They manage how we find websites and how devices get their network settings.

Port 53 (DNS): Converts names like google.com into IP addresses. It is vulnerable to DNS Cache Poisoning and Amplification DDoS attacks.

Port 80 (HTTP): Standard, unencrypted web browsing. It is the primary target for SQL Injection, Cross-Site Scripting (XSS), and Man-in-the-Middle (MiTM) attacks.

Port 443 (HTTPS): Secure, encrypted web browsing. While much safer, it has been targeted by famous exploits like Heartbleed and POODLE.

Port 67/68 (DHCP): Automatically assigns IP addresses to devices. It is vulnerable to DHCP Starvation (running out of IPs) and Rogue DHCP server spoofs.

Port 123 (NTP): Synchronizes time across network devices. It is frequently used in NTP Amplification to launch massive DDoS attacks.

Port 161/162 (SNMP): Used to manage network hardware like routers. Vulnerabilities include Information disclosure via "community strings" that are easy to guess.

Port 8080 (HTTP-Alt): Often used for web proxies or internal testing. It shares the same vulnerabilities as Port 80 but is sometimes overlooked by security filters.

3. Email Services

Email protocols are the "middlemen" for your messages. They are constantly probed for stolen credentials and used to spread spam.

Port 25 (SMTP): Used for sending and relaying emails. It is often exploited for Spam relaying and Email Spoofing.

Port 110 (POP3): An old way to download emails. Because it is unencrypted, it is a target for Credential sniffing.

Port 143 (IMAP): Modern email syncing. It is a constant target for Brute-force attacks to gain access to private mailboxes.

Port 465 (SMTPS): Secure SMTP over SSL. Its vulnerabilities usually involve Outdated cipher suites or old versions of SSL.

Port 587 (SMTP Submission): The modern secure way to send mail. Vulnerable primarily to Authentication attacks and social engineering.

Port 993 (IMAPS): Encrypted IMAP. Like Port 143, it is targeted via Credential theft rather than breaking the encryption itself.

Port 995 (POP3S): Encrypted POP3. It can be vulnerable if the server uses Outdated SSL/TLS implementations.

4. Databases & Internal Windows Services

These ports should almost never be exposed to the public internet. If they are, they represent an open invitation for data breaches.

Port 135 (RPC): Used for Windows Remote Procedure Calls. It is a major vector for Lateral movement (moving from one hacked PC to another) in a network.

Port 137-139 (NetBIOS): Legacy Windows sharing. Vulnerable to Information leakage and older SMB exploits.

Port 445 (SMB): Modern Windows file sharing. This was the entry point for the WannaCry (EternalBlue) ransomware attack.

Port 389 (LDAP): Used for centralized user login. It is vulnerable to Injection attacks and "Anonymous Binds" that allow hackers to see user lists.

Port 1433 (MSSQL): Microsoft’s database engine. It is a high-value target for SQL Injection and brute-forcing the "SA" (System Admin) account.

Port 3306 (MySQL): The most popular open-source database. It is frequently targeted for Remote code execution if passwords are weak.

Port 5432 (PostgreSQL): Another common database. Vulnerabilities often involve Misconfigured permissions or default password exploits.

Port 6379 (Redis): A super-fast data store. Its biggest danger is Unauthenticated RCE, where a hacker can take full control because no password was set.